# Node security: what you can and can't do

The keys, who can touch your money, the rules that keep it safe, and the mistakes that lose it. For the Altradits LND node on the Mac, and any LND node you practise on.

## The one idea to remember

> **Whoever can run commands with a spending key can move your money.** Everything else (the dashboard, the website, the tunnel, your Lightning peers) is built so it can look or receive, but never spend. Keep spending power in one place: this Mac, in your hands, only while you are using it.

## The keys (macaroons)

LND doesn't use passwords for commands. Each command carries a **macaroon**, a key file that says what it may do. They live in `data/chain/bitcoin/mainnet/` inside the node folder.

| Key | Can do | Can't do | Keep it |
|---|---|---|---|
| **admin.macaroon** | Everything: send, withdraw, open and close channels, make new keys | Nothing is off limits | Delete after use. LND makes a new one at each restart |
| **readonly.macaroon** | Read balances, channels, payments, invoices | Spend, create invoices, change anything | For daily checks (ln-ro) and the dashboard |
| **invoice.macaroon** | Create and look up invoices | Spend anything | The kind the website uses to get paid |
| **router.macaroon** | Send Lightning payments | On-chain sends, new keys | Delete if unused: it can drain the channel |
| **Your own (bakemacaroon)** | Exactly the permissions you choose | Anything you left out | Bake narrow keys for each job |

A key is only as safe as the file. Anyone who copies a spending key and can reach the node can spend. Never paste a key's hex into a chat, website or document.

## Who can do what on this setup

| Who | What they can do | Why not more |
|---|---|---|
| **You, at the Mac, with the admin key** | Everything | This is where spending lives |
| **You, with ln-ro** | Look at everything | The read-only key can't spend |
| **The altradits.com website** | Create invoices, read balances | It holds only invoice and read-only keys |
| **The admin dashboard** | Show numbers, make top-up invoices | The spending key never leaves the Mac |
| **Anyone on the internet** | Nothing | LND listens only on 127.0.0.1; the tunnel only lets the website's server in |
| **Your Lightning peer** | Route payments, close the channel | Can't take your side of the channel |
| **Someone logged into your Mac account** | Almost everything | They can read the unlock file and run lncli |

The last row is why your Mac login is the real vault door: strong password, FileVault on, screen lock, and no one else using your account.

## What you can and can't do with your money

| You can | You can't |
|---|---|
| Send up to your side of the channel minus the reserve | Send the reserve (1% of the channel) while it is open |
| Receive up to the other side's balance | Receive more than that without more inbound liquidity |
| Retry a failed payment: a failed payment never loses money | Reverse a payment that SUCCEEDED |
| Close a channel cooperatively and get your side back on-chain in about an hour | Get force-closed funds back quickly: they are locked for a delay of up to 2 weeks |
| Restore funds with the seed plus channel.backup | Restore with the seed alone and keep your channels: channel funds need channel.backup |

## Good practice, every time

• **Check before you pay.** Decode every invoice with the read-only key (decodepayreq): amount, network, expiry, recipient.

• **Cap the fee.** Always pay with `--fee_limit`, and `--timeout 5m` for hard routes.

• **Type yes in full.** lncli asks for confirmation; read the amount on that screen first.

• **Delete the admin key** when you finish, and after every restart.

• **Back up channel.backup** to somewhere other than the Mac after every channel opens or closes.

• **Practise on regtest or signet first** (Polar is installed). Only repeat on mainnet once a drill is boring.

• **Update carefully.** Download LND only from its GitHub releases and verify the signature and checksum.

• **Keep the Mac awake, plugged in and online** while the node takes payments.

• **One node per data folder.** If lnd says *address already in use*, a copy is already running; don't start another.

## Never do these

> **Never run the same wallet and channels on two machines at once**, and never restore an old copy of the node's data folder. An old channel state looks like cheating to your peer, who can take **the whole channel** as a penalty. To move the node, stop the old one for good first, or restore from seed plus channel.backup, which closes the channels safely.

• Never type the 24-word seed into a computer, phone, chat or photo. Paper only.

• Never share the wallet password, the unlock file or any macaroon hex.

• Never put your mainnet seed on a practice machine.

• Never pay an invoice you didn't decode, or one someone else sent you "to test".

• Never open the node's ports (9735, 10009, 8080) to the internet. Peers reach you over Tor; the website through the locked tunnel.

• Never force close unless the peer is gone for good: cooperative closes are cheaper and faster.

## If something goes wrong

| Situation | Do this |
|---|---|
| **A payment shows IN_FLIGHT** | Wait. Don't pay a second invoice until it says SUCCEEDED or FAILED. |
| **You think someone got into the Mac** | Stop LND (pkill -TERM -x lnd), change the Mac password, then withdraw to a wallet you trust. |
| **A key file leaked** | Stop using it. Ask for help to rotate keys (lncli bakemacaroon, and deleting old macaroon IDs). |
| **The Mac is lost or dead** | On a new machine: install LND, lncli create, choose y, enter the seed, give it channel.backup. |
| **The node won't start** | Read the last lines of ~/.lnd-altradits/logs/launchd-lnd.log before changing anything. |